NATCHANON PRECHATAVANICH

張清鬆



Email: natchanon.precha@gmail.com, f11k43026@ntu.edu.tw
Website: prechatavanich.github.io
linkedin.com/in/natchanon-prechatavanich/
Location: Taipei, Taiwan

Education

National Taiwan University, Taipei, Taiwan 2022–2026

GSAT, Ph.D. Program for Semiconductor Device and Hetero-integration (DMHI)

- Advisor: Professor. Chao-Hsin Wu, Ph.D.
- GPAX: 4.25/4.30 (Ranking: 1/11)

Chulalongkorn University, Bangkok, Thailand 2018–2022

Bachelor of Electrical Engineering (EE)

- Advisor: Professor. Songphol Kanjanachuchai, Ph.D.
- GPAX: 3.81/4.00 (1st Class Honors)

Academic Achievement

- Journal Papers
  - **N. Prechatavanich** et al., “Sidewall optical leakage suppression via sidewall reflector in sub-20 μm AlGaInP micro-resonant cavity LEDs,” *J. Lightw. Technol.*, early access, 2026, doi: 10.1109/JLT.2026.3739778.
  - **N. Prechatavanich** et al., “Sidewall optical leakage and scaling limits in sub-20 μm AlGaInP micro-resonant cavity LEDs,” *APL Photonics*, vol. 11, no. 5, 2026, doi: 10.1063/5.0321001.
  - **N. Prechatavanich** et al., “Sidewall defect suppression of 620 nm AlGaInP-based red μLED devices using HfO_2 ALD passivation,” *Opt. Lett.*, vol. 50, no. 2, p. 313, 2025, doi: 10.1364/OL.543122.
  - Z.-A. Lin, **N. Prechatavanich**, et al., “Temperature, shape, and size dependence of sidewall defect recombination in AlGaInP red micro-LEDs,” *IEEE Trans. Electron Devices*, to be published.
  - M.-J. Wu, C.-K. Yee, **N. Prechatavanich**, et al., “Cryogenic insights into the defect-induced non-radiative recombination mechanisms of AlGaInP red micro-LEDs,” *ACS Photonics*, to be published.
  - T. Sutheebanjerd, **N. Prechatavanich**, et al., “Suppression of angular color shift in AlGaInP-based red resonant cavity light-emitting diodes via structural optimization,” *J. Phys. D: Appl. Phys.*, vol. 58, no. 45, Art. no. 455103, 2025, doi: 10.1088/1361-6463/ae16b8.
  - M. Kumar, …, **N. Prechatavanich**, et al., “4-to-2 optoelectronic encoders utilizing quantum well heterojunction phototransistors on the light-emitting transistor platform,” *IEEE Electron Device Lett.*, vol. 46, no. 9, pp. 1565–1568, 2025, doi: 10.1109/LED.2025.3589891.

- C.-E. Chen, **N. Prechatavanich**, et al., "Edge breakdown suppression in avalanche photodiodes using an attached step guard ring," *IEEE Trans. Electron Devices*, vol. 71, no. 5, pp. 3062–3068, 2024, doi: 10.1109/TED.2024.3378222.

➢ Conference Papers
- **N. Prechatavanich** et al., "Transient response of AlGaInP micro-RCLEDs for visible light communication," in *Compound Semiconductor Week (CSW)*, Kumamoto, Japan, May 2026.
- **N. Prechatavanich** et al., "620 nm resonant-cavity LEDs for high color purity and light output power enhancement," in *Conf. Lasers Electro-Opt./Pacific Rim (CLEO-PR)*, Incheon, South Korea, Aug. 2024.

Patents

➢ C.-H. Wu, …, **N. Prechatavanich**, et al., "Infrared full color image sensor," U.S. Patent Appl. Pub. US 2026/0090183 A1, Mar. 26, 2026 (filed Oct. 30, 2024).

➢ C.-H. Wu, C.-E. Chen, and **N. Prechatavanich**, "Guard ring structure and component structure," U.S. Patent Appl. Pub. US 2024/0405131 A1, Dec. 5, 2024 (filed Aug. 7, 2023).

Research Projects

➢ Digital-output infrared readout circuit development and chip integration (NCSIST project)
- Infrared photodiode simulation and design optimization 2025-Present

➢ Efficiency enhancement of red micro-LEDs (NTU–AUO industry collaboration)
- Red RC micro-LED efficiency enhancement 2024-Present
- Sidewall passivation for red micro-LED efficiency 2022–2024

➢ NTU–PSMC industry collaboration
- Quantum dot upconversion compatible with CIS technology 2023-2024

Professional Experience

➢ Taiwan Semiconductor Manufacturing Company Limited (TSMC), Taiwan.
- 2024 DNA internship program, CIS Technology Platform, Fab 12, Hsinchu.

➢ Inronics Limited, Thailand.
- 2021 Internship program, Intern Engineer.

Academic Visit / International Exchange

➢ Université Grenoble Alpes (UGA) & Institut Néel (CNRS Alpes Campus), Grenoble, France.
- 2026 NTU–UGA–UT Trilateral Workshop & laboratory visit, April 27–28, 2026, under the NTU–UGA–UT Trilateral Center program

SKILLS & ABILITIES

➢ **Basic computer languages**
- Python | MATLAB | C++ | VHDL (FPGA)

➢ **Device simulation software**
- Victory Device Silvaco | RSoft Synopsys | Meta Optics Design Synopsys

➢ **Languages**
- Thai: Native | English: TOEIC 905 | Mandarin: TOCFL B2